

Hassanpreet Dhaliwal

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EDUCATION

- Texas Tech University, Department of Geosciences (Atmospheric Science Division)** *Lubbock, TX, USA*
PhD, Geosciences 08/2026
Thesis: Drylands amidst the Weather-Climate Continuum: Boundary Layer Perspectives on Aerosol Mixing Processes and Land-Atmospheric Interactions
Advisor: Dr. Sandip Pal, Texas Tech University
- Texas Tech University, Department of Geosciences (Atmospheric Science Division)** *Lubbock, TX, USA*
Master of Science, Atmospheric Science 12/2022
Thesis: Impact of coal fly ash and soot particles on human lung cells
Advisor: Dr. Karin Ardon-Dryer, Texas Tech University
- Panjab University, Department of Physics** *Chandigarh, India*
Master of Science, Physics (Honors) 05/2019

RESEARCH EXPERIENCE

- Texas Tech University** *Lubbock, TX, USA*
Research Assistant 01/2023 – 07/2026
- Created a novel algorithm to detect soil moisture (SM) freezing and thaw transitions by analyzing over five years of high-resolution data to analyze sub-seasonal land-atmosphere interactions
 - Analyzed National Land Cover Database (NLCD) land-use and land-cover products to characterize how land-atmosphere interactions vary across different land-cover regimes in the U.S. Southwest.
 - Developed statistically robust methods in Python (NumPy, Pandas, xarray, SciPy, Cartopy, matplotlib) for analysis of five years of collocated satellite-retrieved (from MODIS, VIIRS) and ground based atmospheric data (AERONET, Mesonet stations, ASOS, AWOS, NCEI) to identify retrieval errors
 - Developed scalable Apache Spark and machine learning enabled pipelines to integrate satellite, lidar, in situ particulate matter, and meteorological datasets, to analyze how much of aerosol optical depth (AOD) resides in the atmospheric boundary layer
 - Managed the AERONET data visualization on the project website for the NASA-sponsored Synergetic Surface-based and Satellite-borne Measurements of Arid region Aerosol and Precipitation initiative
 - Deployed and operated a Texas Tech scanning Doppler lidar, low-cost HOBO temperature sensors network, and an Aerosol Robotic Network (AERONET) sun photometer
- Unidata, University Corporation of Atmospheric Science (UCAR)** *Boulder, CO, USA*
Software Research Intern 05/2022 – 07/2022
- Developed planetary boundary layer depth retrieval algorithm from COSMIC-I radio occultation (RO) Sounding data (sub-setting via ncdump) and NASA DISCOVER AQ-2013 data via the Integrated Data Viewer (IDV), a python and java-based application
- NASA TRACER Campaign: Forecasting Team** *Remote*
Air Quality (AQ) Forecaster 09/2021 & 07/2022
- Forecasted AQ, ozone, and aerosol transport via multiple models runs and outputs including Hysplit back trajectories during six IOP Projects in Houston region during NASA's Tracking Aerosol Convection Interactions Experiment (TRACER - AQ)
- Texas Tech University** *Lubbock, TX, USA*
Research Assistant and Lab Manager 08/2022 – 12/2022
- Planned experiments to analyze in-vitro toxicity of particulate matter (PM) including soot, coal fly ash, and African dust on human lung epithelial (A549) cells in aa Bio-safety Level 3 (BSL-3) Lab

Department of Physics, Panjab University

Physics Laboratory Researcher

Chandigarh, India

05/2022 – 06/2022

- Developed C++ code for Monte Carlo simulations of nuclear radioactivity, fluid convection modeled with the Lorenz system, and artificial satellite orbital dynamics

PUBLICATIONS

Dhaliwal, H. and Pal, S. (2026). *Weather specific evaluation of satellite retrieved aerosol optical depth over drylands of the Southwest United States*. Atmospheric Research. <https://doi.org/10.1016/j.atmosres.2026.108932>

Dhaliwal, H., Pal, S. & Weckwerth, T. (2026). *Reversals in Soil Moisture-Temperature Coupling under Contrasting Vapor Pressure Deficits over a Dryland Ecosystem* [Under revision]. Journal of Geophysical Research: Atmospheres.

Pal, S., Hamel, M., **Dhaliwal, H.**, Harr, D., Danzig, T., Lee, T. R., Das, D., Menon, K., Prince, N. E., Asel, M., & Burgett, W. (2026). *How a nocturnal cold front amplified wildfire impacts on near-surface air quality downwind of the second largest U.S. wildfire*. Atmospheric Research. <https://doi.org/10.1016/j.atmosres.2026.108841>

Danzig, T., Pal, S., Medley, Z., **Dhaliwal, H.**, Hamel, M., Sorensen, M., Das, D., Menon, K. V., Lee, T. R., & Conder, M. (2025) *Impact of advection on the urban heat of a small-sized city: The Urban Heat Island Experiment around Lubbock, Texas (U-HEAT)*. Urban Climate. <https://doi.org/10.1016/j.uclim.2025.102714>

Dhaliwal, H. (2022). *Effect of soot and coal fly ash particles on human lung cells* [Master's thesis, Texas Tech University]. TTU Electronic Theses and Dissertations. <https://hdl.handle.net/2346/96989>

Pal et al. (with **Dhaliwal, H.**, as co-author). 2026. *On the Impact of Gusts on Dusts during Dryline Passages over Drylands: A Lidar-based Investigation of Boundary Layer Aerosol Mixing* [Invited, in preparation.], Environmental Sciences-Atmospheres.

Dhaliwal, H. (2026). *Drylands through a synoptic lens: Boundary layer perspectives on land-atmosphere and aerosol interactions* [PhD dissertation, Texas Tech University]. TTU Electronic Theses and Dissertations. (Link available soon)

RESEARCH GRANTS & PROPOSALS

NASA Postdoctoral Program (NPP) 2026, Jet Propulsion Laboratory (JPL)

Research Proposal Title: A Novel Framework for Mapping Human Exposure to Speciated Fine Particulate Matter in Regions with Sparse Surface Monitor Coverage

Host: Dr. David Diner, JPL

Status: Proposal Rated “Selectable,” March 2026 Round; Resubmission in preparation for November 2026

University Corporation of Atmospheric Science (UCAR) 2026 Climate & Global Change Postdoctoral Fellowship

Host: Dr. Shima Sham, National Corporation of Atmospheric Science (NCAR)

Status: In preparation

2021 Graduate Student Research Support by Graduate school, TTU

Secured competitive university-wide research funding to purchase laboratory equipment for master's thesis research

Status: Secured

TECHNICAL SKILLS

Programming and Data Analysis: Python (NumPy, Pandas, xarray, SciPy, Matplotlib, Cartopy), PySpark, C++, Bash, NetCDF, High-Performance Computing (HPC) environments for large atmospheric datasets

Atmospheric Models and Model Data: Weather Research and Forecasting (WRF) model; analysis of WRF and High-Resolution Rapid Refresh (HRRR) model output

Remote Sensing and Atmospheric Data: VIIRS, MODIS, AERONET, PurpleAir, EPA monitoring networks, ASOS/AWOS, Mesonet, lidar and radiosonde data

Scientific and Geospatial Software: QGIS, ArcGIS, ENVI, NCAR SOLO3, Unidata IDV, NASA Worldview, NASA Giovanni

Scientific Instrumentation: Doppler lidar, AERONET sun photometer, HOBO sensor networks, PurpleAir sensors, radiosondes

TEACHING EXPERIENCE

Texas Tech University
Teaching Assistant

Lubbock, TX, USA
01/2021 – 05/2022

- Independently taught and designed the undergraduate Introduction to Atmospheric Science course for ~90 students per semester while implementing a mix of formal and informal teaching methods including hands-on experiments, and discussions to enhance student engagement

CONFERENCE PRESENTATIONS

Dhangar, N.G., Pal, S., **Dhaliwal, H. K.**, and Das, D., **(2025)**. Exploring Land-Atmosphere Feedback Processes to Diagnose Numerical Weather Prediction Model Uncertainties in Predicting Mesoscale Weather Events Over Southwest US, submitted to 21st Conference on Mesoscale Processes, 23-26 June 2025, Boise, ID, USA.

Dhaliwal, H. K., & Pal, S. **(2025, January)**. Spatiotemporal Variability of AOD over the Dryland Ecosystem of the US Southwest using a Lens of Synoptic-scale Forcing. In 105th AMS Annual Meeting. AMS.

Pal, S., Hamel, M. B., Danzig, T. B., **Dhaliwal, H. K.**, DAS, D., Menon, K., ... & Lee, T. R. **(2025, January)**. Exploring Vertical and Horizontal Transport of Aerosol Plumes during the Texas Panhandle Wildfire Event of Feb 2024. In 105th AMS Annual Meeting. AMS.

Dhaliwal, H. K., and Pal, Sandip. **(2024, September)**. A Novel Approach for Weather-specific Evaluation of Satellite Retrieved AOD Using AERONET Observations. 2024 AERONET Science and Application Exchange, College Park, MD, Sep 2024.

Dhaliwal, H. K., Pal, S., and Hirth, B. **(2024, January)**. A Comprehensive Analysis of Aerosol Optical Depth Changes on Diurnal, Synoptic, and Seasonal Timescales over the Texas Panhandle: Insights into Aerosol Boundary Layer Interactions in Arid Regions. 2024 American Meteorological Society Annual Meeting, Baltimore, MD, Jan 2024.

Danzig, T., Pal, S., Medley, Z. M., **Dhaliwal, H. K.**, Hamel, M., Lee, T. R., and Conder, M. R., **(2024, January)**. Urban Heat Experiment Around Lubbock, Texas (U-HEAT): An Exploration of Urban Heat Islands and Associated Advection: 2024 American Meteorological Society Annual Meeting, Baltimore, MD, Jan 2024.

Pal, S., Bruning, E., **Dhaliwal, H. K.**, Hirth, B., Ardon-Dryer, K., Schroeder, J., and Weiss, C. **(2024, January)**. Preliminary findings from the Synergetic Surface-based and Satellite-borne Measurements of Arid-region Aerosol and Precipitation (S3-MAAP): 2024 American Meteorological Society Annual Meeting, Baltimore, MD, Jan 2024.

Pal, S., Bruning, E., **Dhaliwal, H. K.**, Hirth, B., Ardon-Dryer, K., Schroeder, J., and Weiss, C. **(2023, December)**. Synergetic Surface-based and Satellite-borne Measurements of Arid-region Aerosol and Precipitation (S3-MAAP): An Overview of Selected Research Highlights. 2023 American Geophysical Union Fall Meeting, San Francisco, CA, Dec 2023.

Dhaliwal, H. K., Pal, S., and Hirth, B. **(2023, December)**. Unraveling Aerosol – Boundary Layer Interactions in Arid Regions: Insights from Aerosol Optical Thickness measurements over the West Texas Panhandle, **2023** American Geophysical Union Fall Meeting, San Francisco, CA, Dec 2023.

Danzig, T., Pal, S., **Dhaliwal, H. K.**, Hamel, M., Medley, Z., Lee, T.R., and Conder, M. **(2023, December)**. Understanding urban heat islands and heat advection around a small city: Urban Heat Experiment Around Lubbock, Texas (U-HEAT), AGU Fall Meeting, San Francisco, CA, Dec 2023.

Dhaliwal, H.K. **(2023, March)**. Synergetic Surface-based and Satellite-borne Measurements of Arid-region Aerosol and Precipitation (S3-MAAP). Paper presented at the 22nd Annual Graduate Student Research Poster Competition, Texas Tech University, Lubbock, TX

Dhaliwal, H. K., Ho, Y., Camron, D., Carter, S. **(2023, January)**. Determining atmospheric boundary layer depth via Integrated Data Viewer (IDV) from atmospheric sounding profile type data. Paper presented at the 22nd Annual Student Conference of AMS, Denver, CO. <https://ams.confex.com/ams/103ANNUAL/meetingapp.cgi/Paper/409472>

Dhaliwal, H. K., Reynolds, R., Goldstein, H., Kanji, Z., Gao, K., Ardon-Dryer, K. **(2022, January)**. Effect of coal fly ash and soot particles on human lung cells. Paper presented at the 102nd Annual Meeting of American Meteorological Society. Virtual. <https://ams.confex.com/ams/102ANNUAL/meetingapp.cgi/Paper/393557>

Dhaliwal, H.K. **(2022, July)**. Determining atmospheric boundary layer depth via Integrated Data Viewer (IDV) from atmospheric sounding profile type data. Paper presented at the Student Symposium. National Center for Atmospheric Research (NCAR) student symposium, Boulder, Colorado.

Ardon-Dryer, K., Ayomide, T., **Dhaliwal, H. K.**, & Plantier, M. **(2021, December)**. Impact of Various African Dust Particles on Human Health. In AGU Fall Meeting Abstracts (Vol. 2021, pp. A32B-10).

Dhaliwal, H.K. **(2019, February)**. Delegate to Indo-French Conference on Multifragmentation, Collective Flow & Sub Threshold particle Production in Heavy Ion Reactions 2019. CEFIPRA by Department of Science & Technology & Renewable Energy. Chandigarh. India.

SELECTED AWARDS AND HONORS

National Science Foundation (NSF) Travel Grant Awarded for presenting research at the American Meteorological Society (AMS) Annual Meeting.	01/2026
NASA Travel Grant Awarded for presenting research at the 2024 AERONET Science and Application Exchange at the University of Maryland, College Park, MD and visit Goddard Space Flight Center, Greenbelt, MD	09/2025
INSPIRE–SHE (Scholarship for Higher Education), DST, Government of India National scholarship awarded to the top 1% of AISSCE performers; provides up to five years of higher-education support, including a funded research internship.	05/2014 – 05/2019
Unidata UCAR Travel Grant Awarded for presenting research at the American Meteorological Society (AMS) Conference 2023.	01/2023
TTU Geosciences Incent Scholarship Departmental acknowledgment of superior academic standing.	08/2022 – 07/2026
Summer Thesis/Dissertation Research Award by Graduate school, TTU Awarded by the Graduate School for excellence in thesis execution and timely completion.	06/2022 – 08/2022

ACADEMIC SERVICE, LEADERSHIP, & OUTREACH

Session Chair, American Meteorological Society (AMS) 105th Annual Meeting	<i>New Orleans, LA, 2025</i>
• Chaired the session "Advancements in Ground-based, Airborne, and Satellite Remote Sensing of Aerosols: Retrieval, Validation, and Applications." In 25th Symposium on Meteorological Observation and Instrumentation. • Submitted and co-developed the official AMS session topic proposal.	
Committee Member (Nominated position), AMS Committee on Measurements	<i>2024 – Present</i>
• Organize symposia, short courses, and town hall sessions at the American Meteorological Society (AMS) Annual Meeting to advance atmospheric measurements and observational sciences. • Develop online content and engagement materials (Instagram and X) to support committee initiatives and outreach efforts.	
Peer Reviewer	<i>2024 – Present</i>
• Reviewer for Physics and Chemistry of the Earth, Geophysical Research Letters and Atmospheric Science Letters.	
Cultural Ambassador, Texas Tech University	<i>2023 – Present</i>
• Served as a representative of Indian culture in outreach events coordinated by the TTU Office of International Affairs to encourage student engagement at local high schools.	
International Student Representative, TTU Student Graduate Assembly (SGA) (2022-23)	<i>08/2022 – 12/2022</i>
• Advocated for the needs and perspectives of international graduate students within the graduate student body.	